R111/A_family_conditional | Evidence boundary

Single new arm; historical comparisons are provisional, not a performance claim.

One new full18 R111 model, not an arm comparison. Fixed source-lineage families; P(method)=P(family)*P(method|family), normalized over the complete legal pool. Each family has two private full-node EncoderLayer self-attention/FFN layers before pooling, not single-query readout after pooling. Root reads only shared h/context and group-mean method tokens (plus family identity); private features do not feed back into root/shared joint/context. One unweighted NLL on normalized log probabilities; no second softmax, donor, teacher or auxiliary objective. Wrong-family private classification gradients are zero, not necessarily AdamW weight-decay/momentum updates. Shared features are not gradient-isolated. Singleton conditional probability is one; its private features may be skipped. All legal joint probabilities compete at inference; no hard family routing or winner-driven forward. Two4090 DDP, one group, global1024/local512; 57600 updates, warmup4320, AdamW1e-4 to1e-6, seed2. Original R84 frozen NEW90k data/cache and legal pools, TRAIN81000/VAL4500/TEST4500 per task. TRAIN-only SBS and per-instance legal-pool Oracle; ALL task-equal, mixed raw cost descriptive only. Native FP64 ind is the main winner, FP32 argmin only compatibility. One strict-first minimum full18 VAL macro_vs_sbs checkpoint for TRAIN/VAL/TEST/LIB, no reselection. Historical R84A/R94B and any R110 summaries are not newly trained, parameter- or FLOP-matched controls. NSS is saved R79 historical weights on exact NEW4500 TEST/native LIB, not new90k retraining. TEST has participated in development. Mechanism approval is not evidence of performance: single-arm historical comparisons remain provisional, with no causal or performance-gain claim. Optional timing is included only from an existing same-lock receipt; no benchmark is launched, and GPU-resident timing is not end-to-end dataset/solver latency.

Historical controls: R84A: included; R94B: included

R110 (not awaited): A_tied_ce: complete_saved_historical; B_private_ce: complete_saved_historical

Optional timing: included_saved_receipt

R111/A_family_conditional | Same checkpoint full TRAIN

TRAIN is diagnostic, not checkpoint selection. ALL is task-equal.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.4054	66.0332	11.389643	11.400702	11.345073	-0.1935	+0.3914	0.4010
TSP	46.1889	32.8617	8.005213	8.023959	7.970215	-0.2336	+0.4391	0.4342
CVRP	47.9210	18.1284	18.043639	18.151987	17.946816	-0.5969	+0.5395	0.7145
ATSP	76.6840	46.9210	1.640145	1.678679	1.633938	-2.2955	+0.3799	0.3826
OVRP	41.6988	34.2420	7.443683	7.453531	7.397676	-0.1321	+0.6219	0.6330
VRPB	80.6222	80.2765	9.359284	9.357264	9.319342	+0.0216	+0.4286	0.4410
VRPL	38.2012	31.5309	12.392543	12.400882	12.344503	-0.0673	+0.3892	0.3960
VRPTW	80.4395	80.2407	18.812493	18.813703	18.757954	-0.0064	+0.2908	0.3049
OVRPTW	83.8864	83.8025	10.118271	10.118640	10.093220	-0.0036	+0.2482	0.2434
OVRPB	88.4370	88.3580	6.363100	6.363230	6.346658	-0.0020	+0.2591	0.2668
OVRPL	42.2778	34.4716	7.443295	7.453219	7.398483	-0.1332	+0.6057	0.6169
VRPBL	79.9025	79.4593	9.403340	9.403115	9.363196	+0.0024	+0.4287	0.4312
VRPBTW	79.6062	79.4914	19.732311	19.733549	19.642712	-0.0063	+0.4561	0.4410
VRPLTW	80.5074	80.3827	18.769052	18.769959	18.714366	-0.0048	+0.2922	0.3071
OVRPBL	88.6247	88.5556	6.359005	6.358857	6.342513	+0.0023	+0.2600	0.2682
OVRPBTW	83.3420	83.2432	10.623707	10.624446	10.586225	-0.0070	+0.3541	0.3287
OVRPLTW	83.8111	83.7321	10.110834	10.111265	10.085768	-0.0043	+0.2485	0.2460
VRPBLTW	79.7679	79.5630	19.768698	19.770814	19.680364	-0.0107	+0.4488	0.4311
OVRPBLTW	83.3790	83.3370	10.624962	10.625544	10.587365	-0.0055	+0.3551	0.3312

R111/A_family_conditional | Fixed36 vs R84A

Original R84A target list; true correct/rescued/lost and false picks are distinct.

Split	Winners	Correct	Rescued	Lost	False	New false	Net cost
TRAIN	83279	597	361	61	521	393	0.000482771
VAL	4625	31	23	3	23	21	0.000431068
TEST	4598	23	14	4	38	29	0.000815976

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.161523	23349	20994	0.000482771	0.005761
VAL	0.028395	1281	1258	0.000431068	0.004475
TEST	0.095062	1291	1214	0.000815976	0.008471

Net cost/regret uses the full18 aligned population, not only target winners.

Raw cost deltas are descriptive; no causal, matched-budget or efficacy conclusion.

R111/A_family_conditional | Fixed36 vs R94B

Original R84A target list; true correct/rescued/lost and false picks are distinct.

Split	Winners	Correct	Rescued	Lost	False	New false	Net cost
TRAIN	83279	597	175	2308	521	387	0.003352794
VAL	4625	31	11	106	23	11	0.000763582
TEST	4598	23	5	99	38	23	0.002425415

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.542455	26696	34605	0.003352794	0.033160
VAL	-0.314815	1596	1851	0.000763582	0.008890
TEST	-0.246914	1582	1782	0.002425415	0.024202

Net cost/regret uses the full18 aligned population, not only target winners.

Raw cost deltas are descriptive; no causal, matched-budget or efficacy conclusion.

R111/A_family_conditional | Saved R110 historical summary

VAL-only rows are provisional; no waiting, rerun, cross-version selection or arm comparison.

Model / split	Status	Top1 (%)	mean_cost	vs_SBS (%)	regret (%)
R110/A_tied_ce VAL	complete_saved_historical	71.1037	11.416779039	-0.192984	0.405093
R110/A_tied_ce TEST	complete_saved_historical	71.3074	11.374706635	-0.186717	0.406464
R110/B_private_ce VAL	complete_saved_historical	70.8963	11.417566031	-0.184524	0.412389
R110/B_private_ce TEST	complete_saved_historical	71.2333	11.374359128	-0.192514	0.399449

R111/A_family_conditional | Saved timing receipt

No benchmark run by this reporter. Scope/exclusions are those of the saved measurement.

Problem	Inputs	Repetitions	Milliseconds / input
TSP	128	3	0.487223
CVRP	128	3	0.467554
ATSP	128	3	0.096535
OVRP	128	3	0.092227
VRPB	128	3	0.092357
VRPL	128	3	0.092204
VRPTW	128	3	0.092065
OVRPTW	128	3	0.091930
OVRPB	128	3	0.092104
OVRPL	128	3	0.091879
VRPBL	128	3	0.091844
VRPBTW	128	3	0.092051
VRPLTW	128	3	0.091769
OVRPBL	128	3	0.092139
OVRPBTW	128	3	0.091883
OVRPLTW	128	3	0.092120
VRPBLTW	128	3	0.091902
OVRPBLTW	128	3	0.092250

Scope: complete_standalone_model_GPU_resident_inputs

Excludes: ['dataset I/O', 'CPU transfer', 'routing solvers']

GPU-resident standalone model, not dataset/solver end-to-end latency.